# Current Affairs 28-08-2026

## Credit - Drishti Ias

# Building Strategic Fuel System

★ ★ ★ ☆ ☆

**For Prelims: Strategic Petroleum Reserves , Viability Gap Funding , Strait of Hormuz , E20 , Current Account Deficit , Oil Marketing Companies**

**For Mains:** Governance challenges in managing strategic reserves and emergency-release mechanisms, India’s energy security and dependence on imported hydrocarbons.

**Source:TH**

## Why in News?

**West Asia-related disruptions have exposed vulnerabilities in India’s energy-security system** , particularly its dependence on imported crude oil, LNG and LPG. This has spotlighted **Phase-II of the Strategic Petroleum Reserves (SPR) programme** , along with the government’s consideration of a reported **decade-long, USD 42 billion expansion of strategic fuel reserves** , signalling a shift towards a comprehensive **Strategic Fuel System** .


## Summary

- **India is moving towards a comprehensive Strategic Fuel System** covering crude oil, LNG and LPG to reduce vulnerabilities arising from import dependence and geopolitical disruptions, particularly in West Asia.
- **The focus must go beyond storage** by developing underground gas storage, pipelines, shipping capacity, diversified energy sources and clear financing and emergency-release mechanisms to ensure fuel can be rapidly delivered during prolonged disruptions.


## What is a Strategic Fuel System?

- **About:** A Strategic Fuel System goes beyond the traditional concept of **Strategic Petroleum Reserves (SPRs).** While SPRs refer exclusively to emergency stockpiles of crude oil maintained by a country to cope with severe supply disruptions, a holistic strategic fuel system encompasses:
  - **Crude Oil Reserves:** Underground rock caverns to store crude.

- **Gas Reserves:** Storage infrastructure for LNG and LPG.
- **Alternative Fuel Integration:** Aggressive blending of **biofuels** (Ethanol, Compressed Biogas) to structurally reduce import dependency.
- **Diversified Sourcing:** Expanding supply chains across multiple geographical zones to minimize geopolitical risk.

- **Current Status of India's Strategic Reserves:** India is building its strategic reserves in a phased manner, managed by **Indian Strategic Petroleum Reserves Limited (ISPRL),** a special purpose vehicle under the Ministry of Petroleum and Natural Gas.
  - **Phase-I (2016–2018):** Built without direct budgetary support, Phase-I established underground **rock caverns with a total capacity of 5.33 MMT across Visakhapatnam (Andhra Pradesh), Mangaluru and Padur (Karnataka)** , providing around 9.5 days of emergency crude-oil supply for national consumption.
  - **Phase-II (Upcoming):** Aims to **add 6.5 MMT of commercial-cum-strategic storage capacity,** comprising 4 MMT at Chandikhol, Odisha, and an additional 2.5 MMT at Padur, Karnataka.
    - It will follow a Public-Private Partnership (PPP) model supported by **Viability Gap Funding (VGF)** capped at 60% of the total project cost.
  - **Expansion into Gas Reserves:** India currently has underground strategic crude-oil storage and two underground LPG caverns, but no operational underground natural-gas storage facility.
    - Exposed by supply chain vulnerabilities stemming from conflicts in the West Asia, India plans to build fuel stockpiles designed to cover about two months of crude and LNG demand, and around **six weeks of LPG (cooking gas) consumption.**
    - This USD 42 billion infrastructure may potentially be financed through small levies on natural gas and cooking gas consumers.
  - **Biofuels & Import Diversification:** India is heavily pushing **E20 (20% ethanol blended petrol)** alongside promoting flex-fuel vehicles.
    - To minimize supply risks, India has aggressively expanded its crude oil sourcing portfolio from 27 countries to 41 countries, and LNG sourcing from 6 to 15 countries.

## Why Does India Need a Comprehensive Strategic Fuel System?

- **High Import Dependency:** India, the **world's 3 rd -largest oil importer and consumer** , imports **over 85% of its crude oil requirements** and nearly **50% of its natural gas needs** . Its **5.33 MMT strategic crude reserves** , currently around **80% full** , provide only limited protection against global supply shocks and geopolitical disruptions.
  - India **imported 60.48 million tonnes of crude oil worth USD 49.66 billion in Q1 FY27** , highlighting its heavy dependence on imported oil and vulnerability to global price and geopolitical shocks.
- **Geopolitical Vulnerabilities:** The vast majority of India's crude oil trade passes through critical, volatile maritime bottlenecks like the **Strait of Hormuz** and the **Red Sea** . Any blockade, piracy, or war in these narrow waterways could immediately sever India's energy lifeline.
  - India's **Phase-I Strategic Petroleum Reserves (SPR)** have a total capacity of **5.33 MMT** , providing only about **9.5 days of crude-oil coverage** , with current utilisation at around **64%** . Even with commercial refinery stocks included, India remains **well below the IEA's recommended 90-day strategic buffer** .

- **Protecting the Balance of Payments (BoP):** Global crude oil price spikes can increase India's import bill, widen the **Current Account Deficit (CAD) and fuel imported inflation.**
    - Strategic reserves can reduce exposure to sudden price shocks and help shield domestic consumers, reducing the need for the government to provide subsidies or require **Oil Marketing Companies (OMCs** ) to absorb losses, thereby easing pressure on public finances.
- **Transition to a Gas-Based Economy:** The government is aggressively pushing to increase natural gas in India's energy mix from the current ~6% to 15% by 2030 to reduce emissions.
    - Building a comprehensive system with LNG surface tanks and depleted underground gas reservoirs is the way to safeguard priority sectors (like fertilizer production and city gas distribution) during a global LNG shortage.

# What are the Key Challenges in Building a Comprehensive Strategic Fuel System?

- **High Cost of Building and Maintaining:** The proposed programme could involve around **USD 42 billion** , covering storage infrastructure and the purchase and maintenance of fuel inventories.
    - The financial burden continues after construction, as reserves must be **financed, maintained and replenished** after emergency releases.
    - Replenishment could become particularly expensive when global **fuel and freight prices are high** during a crisis.
- **Delays in Expanding Strategic Storage:** The expansion of India's strategic storage capacity faces **project-execution challenges** .
    - Under Phase-II, the Chandikhol project has experienced delays due to **land-acquisition requirements and finalisation of commercial PPP frameworks** , highlighting the difficulty of converting planned strategic capacity into operational reserves.
- **Lack of Underground Natural-Gas Storage:** India has underground strategic crude-oil storage and LPG caverns but **no operational underground natural-gas storage facility** . Developing this capacity would require suitable depleted reservoirs or salt caverns and extensive geological, engineering and pipeline assessments.
- **Geological and Technical Site Constraints:** Underground gas storage cannot be developed merely because a depleted field is available. Sites must be assessed for **cap-rock integrity, reservoir characteristics, pressure behaviour, existing wells, cushion-gas requirements and pipeline connectivity** .
    - Similarly, salt caverns require suitable **depth, thickness, purity, geometry, groundwater conditions and mechanical properties** .
- **Storage–Deliverability Gap:** A strategic reserve is useful only if the stored fuel can be **withdrawn and transported to consumers at the required rate** .
    - LNG storage requires adequate **regasification and downstream pipeline connectivity** , while LPG caverns need links with **bottling and distribution systems** . Thus, storage must function as part of an integrated supply network.
- **Maritime Chokepoint Vulnerability:** India's dependence on maritime energy routes leaves its fuel supply vulnerable to disruptions at strategic chokepoints such as the **Strait of Hormuz** .
    - Therefore, even large domestic reserves cannot provide complete energy security if replacement cargoes cannot reach Indian ports during a prolonged crisis.
- **Fuel-Specific Storage Challenges:** The three fuels require different storage technologies and

infrastructure. **Crude-oil storage is the most mature** , while scaling underground LPG storage to the proposed **4 MT** would require new caverns, terminals, pipelines and bottling infrastructure.

  - LNG presents a separate challenge because it requires **cryogenic storage at around –162°C** , while underground natural-gas storage requires suitable geological formations and supporting infrastructure.

- **Storage Capacity and Usable Reserves:** A facility's physical capacity does not necessarily represent the amount of fuel immediately available during a crisis.
  - What matters is the **actual inventory and withdrawal rate** , along with the ability to transport the fuel to consumers.
  - Therefore, India's strategic resilience should be assessed in terms of **deliverable fuel availability** , not merely total storage capacity.

## What Measures Can Strengthen India's Strategic Fuel System?

- **Expand Underground Gas Storage:** India should develop **underground natural-gas storage** using suitable depleted oil and gas reservoirs and explore salt-cavern options where geological conditions permit. This would complement surface LNG storage and provide greater flexibility during prolonged supply disruptions.
- **Increase Salt-Cavern Capacity:** Salt caverns can provide **faster injection and withdrawal** , making them useful for short-duration supply balancing. India's salt-bearing formations, particularly in Rajasthan, can be assessed for suitable projects.
- **Adopt Commercial-Cum-Strategic Models:** Greater use of **PPP and commercial-cum-strategic models** can help mobilise private capital while ensuring that the government retains access to storage and inventories during emergencies. Clear rules on financing, ownership and replenishment are essential.
- **Strengthen Downstream Connectivity:** India should expand **pipeline connectivity, regasification capacity and distribution infrastructure** so that stored fuel can be rapidly delivered to refineries, bottling plants, industrial clusters and other priority consumers.
- **Diversify Energy Supply and Shipping:** India should continue diversifying its import sources and shipping capacity to reduce exposure to individual suppliers and maritime chokepoints. Strengthening domestic shipping capabilities and expanding sourcing partnerships with countries such as Algeria and the US would enhance supply chain resilience.
- **International Cooperation:** India can leverage initiatives such as the **International Solar Alliance (ISA)** to deepen energy cooperation, technology sharing and joint research with partner countries, particularly in **energy storage, clean fuels and resilient energy infrastructure** .

## Conclusion

India should adopt an **integrated strategic-energy framework** by expanding crude, LNG and LPG reserves alongside **underground gas storage, pipelines, shipping and diversified import sources** . Clear mechanisms for **financing, emergency release and replenishment** are also essential. The success of the system will ultimately depend on India’s ability to **rapidly access, transport and deliver fuel during prolonged supply disruptions** .

***Drishti Mains Question:***

The effectiveness of a strategic fuel reserve depends more on deliverability than on storage capacity. Discuss.

# Frequently Asked Questions (FAQs)

**1. What is India's Phase-I Strategic Petroleum Reserve capacity?**
Phase-I provides **5.33 MMT of underground crude-oil storage** at Visakhapatnam, Mangaluru and Padur, equivalent to about **9.5 days of emergency supply** .

**2. What does Phase-II of India's SPR programme propose?**
Phase-II proposes **6.5 MMT of commercial-cum-strategic storage** at Chandikhol and Padur through a **PPP model** , with VGF capped at 60% of project cost.

**3. What is the major gap in India's gas-storage infrastructure?**
India has underground crude-oil storage and LPG caverns but **no operational underground natural-gas storage facility** , making gas storage a major area for expansion.

**4. Why is pipeline connectivity important for strategic reserves?**
Storage alone is insufficient because fuel must be **withdrawn and transported rapidly** ; LNG requires regasification and downstream pipelines, while LPG storage needs links with bottling and distribution networks.

**5. What is the key principle behind India's proposed Strategic Fuel System?**
The system should integrate **storage, shipping, pipelines, diversified imports, emergency release and replenishment** , ensuring that reserves are actually deliverable during prolonged disruptions.

[Watch Video on YouTube: ▶ **https://www.youtube.com/embed/o7BOPqesJHk** ]

## UPSC Civil Services Examination, Previous Year Question (PYQ)

### *Prelims*

**Q. Consider the following statements: (2019)**

1. Coal sector was nationalized by the Government of India under Indira Gandhi.
2. Now, coal blocks are allocated on lottery basis.
3. Till recently, India imported coal to meet the shortages of domestic supply, but now India is self-sufficient in coal production.

**Which of the statements given above is/are correct?**

(a) 1 only

(b) 2 and 3 only

(c) 3 only

(d) 1, 2 and 3

**Ans: (a)**

**Q. With reference to the Indian Renewable Energy Development Agency Limited (IREDA), which of the following statements is/are correct? (2015)**

1. It is a Public Limited Government Company.
2. It is a Non-Banking Financial Company.

**Select the correct answer using the code given below:**

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

**Ans: (c)**

---

### *Mains*

**Q.** "Access to affordable, reliable, sustainable and modern energy is the sine qua non to achieve Sustainable Development Goals (SDGs)".Comment on the progress made in India in this regard. **(2018)**

---

# Streamlining India's R&D Funding

★ ★ ★ ☆ ☆

**Source: TH**

# Why in News?

A **NITI Aayog report on 'Ease of Doing R&D in India'** has highlighted significant gaps in India's research funding ecosystem.

- The report notes the need for a **unified system to track who receives research funding, from which agency, for what purpose and whether similar research has already been funded** .

**Click here to Read:** **Key Highlights of the NITI Aayog Reports on the "Ease of Doing Research & Development in India**

# What are the Challenges Facing India's R&D Funding Ecosystem?

## Structural and Financial Challenges

- **Low Gross Expenditure on R&D (GERD):** India's total R&D investment has stagnated **between 0.6% and 0.8% of GDP,** drastically lagging global innovation leaders like Israel (~5%), the US (3.5%), and China (2.4%).
- **Extreme Fund Concentration:** Funding is heavily skewed toward premier, established institutions. Over 80% of funding from the **Anusandhan National Research Foundation (ANRF)** goes to the IIT system, starving state universities and smaller research bodies of crucial capital.
- **Weak Private Sector Participation:** Historically, the system has relied almost entirely on public sources. While private industry spending has recently crossed the 50% mark, the ecosystem still lacks the robust venture capital, philanthropic, and institutional business backing seen in advanced economies (where the private sector routinely contributes >70%).
- **Human Capital Deficit:** India has approximately 262 full-time researchers per million people, compared to nearly 5,000 in the US and UK.
    - This shortage is exacerbated by scientists having to spend extensive time navigating administrative red tape rather than conducting research.

## Administrative and Data Bottlenecks

- **Red Tape and Delays:** Researchers face a massive administrative burden driven by a "trust deficit." Applications are repetitive, disbursements **can take 3 to 6 months post-approval** , and rigid year-end financial rules often force unspent funds to be returned—disrupting ongoing work and delaying stipends.
- **Grant Duplication:** Multiple central agencies **(like DST , DBT , CSIR , and ICMR )** operate in silos.
    - This makes it incredibly difficult to identify overlapping work, costing the ecosystem an estimated thousands of crores annually in duplicate grants for similar research.
- **Fragmented, Inconsistent Data:** Existing grant information is scattered across dozens of separate databases and is often entered as unstructured free text.
    - An automated system cannot easily recognize that **"IISc, Bangalore" and "Indian Institute of Science, Bengaluru" are the same entity.**
- **Information Blindspots:** There is **no centralized dashboard or system to tell funders,**

researchers, or the public who is receiving funding, from whom, for what exact purpose, and whether similar funding exists elsewhere.

## Global Practices in R&D Funding

- **European Union:** The **CORDIS** platform provides a central database of EU-funded research and innovation projects, including project details, results, publications and participating organisations.
- **United Kingdom:** The **Gateway to Research** model provides a national portal through which information on publicly funded research can be accessed and tracked.
- **Global Model: Crossref's Grant Linking System** assigns **Persistent Identifiers (PIDs)** to grants and links them with researchers, institutions and research outputs such as papers and datasets. More than **2 lakh grants** have now been registered globally.
- **Common Identifiers:** These systems use identifiers such as a funder ID (for the agency), a ROR ID (for the institution), an ORCID (for the individual researcher), and a grant DOI (for the specific grant itself) making funding data machine-readable and interoperable.
  - This system allows analysts to detect overlaps and monitor national priorities easily. For instance, a 2013 analysis in the US using automated text-matching on federal grants revealed that duplicate funding was costing the government nearly USD 70 million.

# What Measures are Needed to Solve India's R&D Funding Problem?

- **Persistent Digital Identifier (PID): .** Every research grant should be assigned a unique, machine-readable PID, similar to a PAN for taxpayers or an IMEI for mobile phones.
  - This PID should be linked to standardised metadata covering the funding agency, institution, researcher, grant amount, duration and research field. Researchers should also be required to include the grant PID in publications and patents, allowing authorities to track funding against research outcomes and assess the effectiveness of public R&D spending.
- **Unified Project Management System (UPMS):** The **UPMS** is a proposed **NITI Aayog platform** to bring public R&D funding under a common system.
  - It aims to **streamline the planning, funding, monitoring and evaluation** of research projects.
  - It would bring information from different **Ministries and funding agencies** onto a unified platform.
  - It can help track **research proposals, funding allocation, project progress and outcomes** .
  - However, for effective tracking of duplicate funding and funding concentration, UPMS would also need **standardised metadata and persistent identifiers (PIDs)** .
- **Global Integration:** Indian funding agencies could **plug directly into existing non-profit ecosystems like Crossref.** This allows for faster implementation using established technical standards and makes Indian research immediately visible on a global scale.
- **Mandate Regional Allocation:** Direct the ANRF to legally ring-fence a specific percentage of its budget exclusively for state universities, smaller colleges, and historically underfunded institutions, actively shifting the 80% concentration away from just the IITs.
  - Rapidly deploy the newly approved **Research, Development, and Innovation (RDI) Fund** . Use it

to offer long-term loans at nil/low interest rates and equity infusions to deep-tech startups and industries, directly de-risking private R&D.

- **Raise GERD Target:** Commit to a phased fiscal expansion to raise the Gross Expenditure on R&D from the stagnant 0.65% to at least 2% of GDP.
- **Social Recognition of Researchers:** Research should be brought into **public consciousness** through initiatives such as **PM Fellowships** , awards and wider public recognition. The government can also encourage corporations to **sponsor researchers annually** in their respective fields.
- **Statutory R&D Commitments:** Similar to the **Fiscal Responsibility and Budget Management (FRBM) framework** , India could consider **statutory commitments for R&D spending** , ensuring sustained and predictable public investment in research rather than relying solely on annual budgetary priorities.

# Conclusion

Revitalising India’s R&D ecosystem requires **better governance, wider funding access and transparent tracking** , not just higher budgets. Adopting global digital standards and enabling more flexible, decentralised funding can reduce inefficiencies, strengthen State universities and **bridge the gap between public research and private innovation** .

***Drishti Mains Question:***

A strong R&D ecosystem requires not only higher expenditure but also greater institutional diversity, administrative flexibility and outcome-based funding. Critically examine the statement in the Indian context.

# Frequently Asked Questions (FAQs)

**1. What is the Unified Project Management System (UPMS)?**
UPMS is a proposed **NITI Aayog platform** to streamline the planning, funding, monitoring and evaluation of public R&D projects across Ministries and funding agencies.

**2. What are Persistent Digital Identifiers (PIDs) in research funding?**
PIDs are **unique, machine-readable identifiers** assigned to research grants and linked with standard metadata such as the funder, researcher, institution, amount and duration.

**3. What is the role of ORCID and ROR in R&D funding?**
**ORCID identifies individual researchers** , while **ROR identifies research organisations** , enabling accurate linking and aggregation of research-funding data across databases.

**4. Why is ANRF significant for India's R&D ecosystem?**
The **Anusandhan National Research Foundation (ANRF)** aims to strengthen and broaden India's research base by supporting research and innovation across universities and institutions beyond traditional funding centres.

**5. What is the significance of the RDI Fund?**
The **Research, Development and Innovation (RDI) Fund** seeks to catalyse private-sector investment in R&D, particularly by supporting **deep-tech startups and innovation-intensive industries** through suitable financial instruments.

[Watch Video on YouTube: ▶ https://www.youtube.com/embed/2v9ql6qD-c8 ]

## UPSC Civil Services Examination, Previous Year Questions (PYQs)

### *Prelims*

**Q. Which of the following statements is/are correct regarding National Innovation Foundation-India (NIF)? (2015)**

1. NIF is an autonomous body of the Department of Science and Technology under the Central Government.
2. NIF is an initiative to strengthen the highly advanced scientific research in India's premier scientific institutions in collaboration with highly advanced foreign scientific institutions.

**Select the correct answer using the code given below:**

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

**Ans: (a)**

### *Mains*

**Q.** What are the research and developmental achievements in applied biotechnology? How will these achievements help to uplift the poorer sections of society? **(2021)**

Q. Scientific research in Indian universities is declining because a career in science is not as attractive as are business professions, engineering or administration, and the universities are becoming consumer-oriented. Critically comment. **(2014)**

---

# 5 Years of e-Shram Portal

★ ★ ★ ☆ ☆

**Source: PIB**

## Why in News?

Recently, the **e-Shram Portal** completed five years, marking a significant milestone in creating an **inclusive social security ecosystem** for India's unorganised workforce.

# What are the Key Facts About e-Shram Portal?

- **Implementation:** Launched in **2021 by the Ministry of Labour and Employment** and developed by the **National Informatics Centre (NIC)** , e-Shram created India's first **Aadhaar-authenticated National Database of Unorganised Workers (NDUW)** .
- **Definition of Unorganised Worker:** An unorganised worker includes a **home-based worker, self-employed worker, or wage worker in the unorganised sector** . It also covers workers in the organised sector who are not members of **EPFO, ESIC, or government employees** .
- **Unique Identification:** The portal provides every registered unorganised worker with a **12-digit Universal Account Number (UAN)** , enabling identification and access to welfare services across the country.
- **Objectives:** e-Shram aims to improve the delivery of social security schemes through **inter-ministerial convergence** , secure data sharing through APIs, and evidence-based policymaking.
- **Portability of Benefits:** The platform facilitates nationwide portability of social security and welfare benefits, particularly benefiting **migrant and construction workers** who frequently move across states.
- **Coverage and Services:** With over **31.89 crore registrations** , e-Shram has evolved into a one-stop digital platform integrating **welfare schemes, employment opportunities, skilling, apprenticeship, and pension-related services** .
- **Eligibility Criteria:** Any **unorganised worker aged 16 years and above** can register on e-Shram, provided the worker is not an **income tax payer or a member of EPFO/ESIC** .
- **Coverage of Workers:** The portal covers diverse categories including **construction workers, migrant workers, gig workers, street vendors, domestic workers, agricultural workers, and other unorganised workers** .
- **Significance:** The portal strengthens India's vision of **Viksit Bharat** by promoting a technology-driven, inclusive social security framework and improving welfare access for the country's vast unorganised workforce.

## Evolution into a One-Stop Solution

- **e-Shram One-Stop Solution:** Launched in **October 2024** , the platform evolved from a worker registry into an integrated service delivery platform providing access to **welfare, employment, skilling, and pension services** .
- **Scheme Integration:** The One-Stop Solution integrates **15 Social Security and Welfare Schemes** , including **Pradhan Mantri Shram Yogi Maan-Dhan Yojana (PM-SYM) , Pradhan Mantri Suraksha Bima Yojana (PMSBY) , Pradhan Mantri Jeevan Jyoti Bima Yojana (PMJJBY) , National Social Assistance Programme (NSAP), One Nation One Ration Card (ONORC) , and Pradhan Mantri Awaas Yojana–Gramin (PMAY-G)** , enabling workers to access multiple benefits through a single platform.
- **Insurance Benefits:** e-Shram enables workers to enrol in **PMSBY and PMJJBY** . PMSBY provides **₹2 lakh accidental insurance cover for death/permanent disability and ₹1 lakh for partial disability** , while PMJJBY provides **₹2 lakh life insurance cover** .
- **Employment and Skill Linkages:** The portal connects workers with the **National Career Service (NCS)** for employment opportunities and the **Skill India Digital Hub** for skilling and apprenticeship opportunities.
- **Pension Support:** Eligible workers can enrol in **Pradhan Mantri Shram Yogi Maan-Dhan (PM-SYM)** , which provides a minimum assured pension of **₹3,000 per month after attaining 60 years of age** .

- **Multilingual Accessibility:** Powered by the **Bhashini platform** , e-Shram introduced multilingual services in **22 languages in 2025** .
- **Support for Migrant Workers:** The portal captures **family details of migrant workers** and facilitates portability of welfare benefits across States.
- **Support for Construction Workers:** e-Shram enables sharing of construction workers' data with States/UTs for registration with **Building and Other Construction Workers (BOCW) Boards** .
- **Coverage of Gig Workers:** The portal supports identification and registration of **gig and platform workers** . Around **11.5 lakh platform workers** have registered on e-Shram.

## Conclusion

The **e-Shram Portal represents a major step towards building an inclusive and technology-driven social security ecosystem for India's unorganised workforce** . By creating a national database, providing a unique identity through UAN, enabling portability of benefits and integrating welfare, employment and skilling services, e-Shram has transformed labour welfare delivery.

***Drishti Mains Question***

The e-Shram Portal has emerged as a critical digital platform for social security inclusion of India's unorganised workforce. Discuss.

## Frequently Asked Questions (FAQs)

**1. What is e-Shram Portal?**
e-Shram is a digital platform launched in 2021 to create India's first Aadhaar-authenticated National Database of Unorganised Workers and provide access to social security services.

**2. Which Ministry launched the e-Shram Portal?**
The e-Shram Portal was launched by the **Ministry of Labour and Employment** in 2021.

**3. What is Universal Account Number (UAN) in e-Shram?**
UAN is a unique **12-digit identification number** provided to every registered unorganised worker.

**4. How many workers are registered on e-Shram?**
More than **31.89 crore workers** have registered on the portal.

**5. Who can register on e-Shram?**
Any unorganised worker aged **16 years or above** , who is not an income tax payer and not an EPFO/ESIC member, can register.

**[Watch Video on YouTube:** ▶ **https://www.youtube.com/embed/ABh4gMmaOd8** **]**

## UPSC Civil Services Examination, Previous Year Question (PYQ)

### *Mains*

**Q.** How globalization has led to the reduction of employment in the formal sector of the Indian economy? Is increased informalization detrimental to the development of the country? **(2016)**

---

# PM Street Vendor’s AtmaNirbhar Nidhi

★ ★ ★ ☆ ☆

**Source: PIB**

## Why in News?

The **PM Street Vendor’s AtmaNirbhar Nidhi (PM SVANidhi) Scheme** has completed one year of its **restructured phase** and expands its focus from working-capital credit to **formal finance, digital inclusion, social security, capacity building and entrepreneurship** .

## What are the Key Facts About the PM Street Vendor’s AtmaNirbhar Nidhi (PM SVANidhi)?

- **About:** PM SVANidhi launched in **June 2020** by **MoHUA** to provide **collateral-free working-capital loans** , promote **digital financial inclusion** , and formalise the urban street-vending ecosystem. Launched during the **Covid-19 pandemic** , it helps street vendors reduce dependence on informal lenders and access the **formal banking system** .
- **Restructured Scheme:** Approved by the **Union Cabinet in August 2025** , the scheme’s lending period has been **extended until 31 st March 2030** , targeting **1.15 crore street vendors** , including **50 lakh new beneficiaries** .
- **Nodal Authority & Governance:** Implemented as a 100% centrally funded **Central Sector Scheme** under the **Ministry of Housing and Urban Affairs (MoHUA)** .
- **Target Beneficiary Base:** Urban, peri-urban, and rural **street vendors, hawkers, thelewalas, and informal service providers** (barbers, cobblers, laundry workers) operating within the limits of **Urban Local Bodies (ULBs)** .
- **Graduated Working Capital Facility:**
  - **1st Tranche:** Collateral-free working capital loan up to **Rs 15,000** with a **12-month repayment tenure** .
  - **2nd Tranche:** Scaled loan up to **Rs 25,000** with an **18-month tenure** upon timely or early

repayment of the initial credit.

  - **3rd Tranche:** Enhanced credit facility up to **Rs 50,000** with a **36-month tenure** , featuring **no prepayment penalty** .

- **Interest Subvention & Incentives:** An **interest subsidy of 7% per annum** credited **quarterly via Direct Benefit Transfer (DBT )** into the beneficiary's bank account for standard (non- **NPA** ) accounts.
  - Monthly **digital transaction cashback up to Rs 100** to encourage formal cashless banking.
- **Credit Card Integration:** Issuance of a **UPI-Linked RuPay Credit Card** with an initial credit limit of Rs **10,000** (extendable to **Rs 30,000** ) and a **validity of 5 years** , accessible following successful repayment of the second loan tranche.
- **Digital Lending Platform:** An end-to-end **Digital Lending Platform (DLP)** has simplified the process from **application and verification to approval and disbursement** .
- **Vendor Migration Module:** Enables seamless transfer of **Letters of Recommendation (LoRs)** when vendors move between **ULBs and Census Towns** , ensuring continuity of benefits.
  - The **City Region Approach** expands the scheme's geographical coverage by including **Census Towns** along with urban areas. It enables more street vendors to enter the formal credit and livelihood-support ecosystem.
- **Institutional Eligibility Criteria:** Vendors possessing a **Certificate of Vending** or **Identity Card** issued by **Urban Local Bodies (ULBs)** .
  - Vendors identified in surveys or left-out vendors holding a verified **Letter of Recommendation (LoR)** issued by the **ULB / Town Vending Committee (TVC)** .
  - Peri-urban and rural vendors operating within municipal territorial jurisdictions backed by an authorized **LoR** .
- **'SVANidhi se Samriddhi': To** elevate the scheme from mere credit provision to a holistic welfare net, the MoHUA launched the **'SVANidhi se Samriddhi '** initiative.
  - **Objective:** To map the socio-economic profile of PM SVANidhi beneficiaries and their families and facilitate their access to 8 core central government welfare schemes.
  - **Schemes Covered: Pradhan Mantri Jeevan Jyoti Bima Yojana (PMJJBY), Pradhan Mantri Suraksha Bima Yojana (PMSBY), PM Jan Dhan Yojana, Registration under Building and other Construction Workers (BoCW), PM Shram Yogi Maandhan Yojana , One Nation One Ration Card (ONORC) , Janani Suraksha Yojana , and PM Matru Vandana Yojana (PMMVY).**
- **Impact:** During the first year of restructuring, 21.86 lakh loans worth Rs 6,294 crore were disbursed to 10.56 lakh new beneficiaries.
  - Since its launch, **PM SVANidhi has expanded access to formal credit and digital finance** , with **78.68 lakh vendors accessing over 1.18 crore loans** , alongside interest subsidies and digital cashback.
  - The **PM SVANidhi Startup Challenge** supports solutions for market access, digital payments, financial services and logistics, while **50 Street Food Hubs** will receive financial assistance of up to **Rs 4 crore each** , with an additional Rs 25 lakh for vending zones.
  - **Lok Kalyan Melas** connect vendors with banks and ULBs, and over **6 lakh street-food vendors** have received **FSSAI training** in food safety and hygiene.

## Frequently Asked Questions (FAQs)

**1. What is PM SVANidhi?**
PM SVANidhi is a **Central Sector Scheme launched on 1 June 2020 by MoHUA** to provide **collateral-free working-capital loans** and promote financial inclusion among street vendors.

**2. What are the three loan tranches under PM SVANidhi?**
The scheme provides loans of up to **₹15,000, ₹25,000 and ₹50,000** in successive tranches, subject to timely or early repayment of the previous loan.

**3. What is SVANidhi se Samriddhi?**
It is an initiative that conducts **socio-economic profiling of beneficiaries and their families** and facilitates access to **eight Central Government welfare schemes** .

**4. How does PM SVANidhi promote digital financial inclusion?**
It incentivises **digital transactions through cashback** and has introduced a **UPI-linked RuPay Credit Card** , with a credit limit extendable up to **₹30,000** .

**5. What is the significance of the restructured PM SVANidhi Scheme?**
The restructured scheme extends lending until **31 March 2030** and targets **1.15 crore street vendors** , including **50 lakh new beneficiaries** , with greater emphasis on credit, social security, entrepreneurship and digital inclusion.

[Watch Video on YouTube: ▶ **https://www.youtube.com/embed/RF5NZvPoEAA** ]

### UPSC Civil Services Examination, Previous Year Question (PYQ)

### *Prelims*

**Q.** Can the vicious cycle of gender inequality, poverty and malnutrition be broken through microfinancing of women SHGs? Explain with examples. **(2021)**

**Q.** How has globalization led to the reduction of employment in the formal sector of the Indian economy? Is increased informalization detrimental to the development of the country? **(2016)**

# Indian Navy Gets 1st Multipurpose Vessel Samarthak

★ ★ ☆ ☆ ☆

**Source: PIB**

The Indian Navy achieved a major milestone in maritime self-reliance with the delivery of **'Samarthak', its first indigenous Multi-Purpose Vessel (MPV).**

- **Builder and Location** : The vessel was constructed and delivered by **Larsen & Toubro Ltd. (L&T)** at its **Kattupalli shipyard** near Chennai.
  - L&T has built two MPVs, **INS Utkarsh and INS Samarthak** , for the Indian Navy. INS Utkarsh was launched in 2025 and is currently undergoing tests and sea trials prior to its final handover.
- **Meaning and Designation** : The ship is named **'Samarthak', meaning 'The Proponent'** , which signifies its core role as a dependable, multi-mission enabler that is not restricted to a single specific task.
- **High Indigenous Content** : Promoting domestic manufacturing, the ship comprises **over 75% indigenous content** , standing as a testament to a self-sustaining defense ecosystem.
- **Primary Function** : It is primarily conceived as a **versatile trial and evaluation platform** for testing next-generation indigenous weapons and sensors under development for the Navy.
- **Diverse Operational Roles** : Beyond weapon testing, it is engineered for a wide spectrum of critical duties, including **maritime surveillance and reconnaissance, offshore patrolling** , and **Humanitarian Assistance and Disaster Relief (HADR)** operations.
- **Advanced Capabilities** : The vessel can launch and recover surface and aerial targets, operate **autonomous/unmanned surface and aerial vehicles (UAVs)** , combat marine pollution, and provide heavy-lift and towing support.
- **Strategic Alignment** : The induction of INS Samarthak is a significant leap in indigenous shipbuilding, serving as a direct reaffirmation of the Government's overarching vision of **Aatmanirbhar Bharat (Self-Reliant India).**

**Read more:** India's First Indigenous Polar Research Vessel

# Growing Health Burden of Salmonella

★ ★ ☆ ☆ ☆

**Source: TH**

A Lancet study estimated 4.9 million **typhoid cases** in India in 2023, highlighting the need to address the broader burden of often undiagnosed non-typhoidal **Salmonella infections** and rising **antimicrobial resistance.**

- **Salmonella infections:** Salmonella, or salmonellosis is an infection caused by *Salmonella* bacteria, broadly classified into:
  - **Typhoidal Salmonella** caused by *Salmonella Typhi* and *Salmonella Paratyphi* , leading to **typhoid and paratyphoid (enteric fever)** , while **Non-typhoidal Salmonella (NTS)** mainly causes **gastroenteritis** and may lead to severe **bloodstream infections** in vulnerable groups.
    - Many **non-typhoidal Salmonella infections** are never tested because mild cases often resolve without medical intervention, making their actual burden difficult to estimate.
- **Transmission:** Typhoid mainly spreads through **faecally contaminated food and water** , while NTS has a broader **animal-food-environment transmission cycle** , involving poultry, eggs, meat and contaminated food.
- **Symptoms: Prolonged fever, abdominal pain, diarrhoea, nausea and vomiting** ; severe cases may cause intestinal bleeding or perforation.
- **Diagnosis: Blood culture** is the main diagnostic test for suspected typhoid and should ideally be performed before antibiotics.
- **Prevention:** Key measures include **safe drinking water, sanitation, hand hygiene, thorough cooking of poultry and eggs, proper refrigeration and preventing cross-contamination** .
- **Vaccination:** The **WHO** **recommends typhoid conjugate vaccines (TCV) from six months of age** .
- **AMR (Antimicrobial Resistance) Crisis:** Fluoroquinolones (a class of antibiotics) resistance is rising significantly in S. Typhi, making treatment more difficult.
- **The 'One Health' Imperative:** Because NTS contamination occurs across farms, slaughterhouses, logistics, and household kitchens, addressing it requires a **' One Health' approach** integrating human, animal, and environmental health strategies.

**Read more:** **Drug Resistant Typhoid**